Teacher Wu Kuan | See It at a Glance #6 — $(a - b)^2$: Don't Drop the Middle Term

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1. The trap question

Is $(a - b)^2 = a^2 - b^2$?

2. The method

- Square a^2; remove two strips $a \times b$.
- The corner b^2 was removed twice → add it back once.
- $(a - b)^2 = a^2 - 2ab + b^2$.

3. Parent check method

Have your child test with $a = 3$, $b = 1$: $(3 - 1)^2 = 4$, while $3^2 - 1^2 = 8$.

Practice

1. Expand $(x - 4)^2$.
2. Expand $(3x - 2)^2$.
3. Use $(100 - 1)^2$ to find 99^2.
4. True or false: $(a - b)^2 = a^2 - b^2$. Test $a = 3$, $b = 1$.
5. $a - b = 3$ and $ab = 4$. Find $a^2 + b^2$.

Answers

1. $x^2 - 8x + 16$
2. $9x^2 - 12x + 4$
3. 9801
4. False (4 vs 8)
5. 17

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